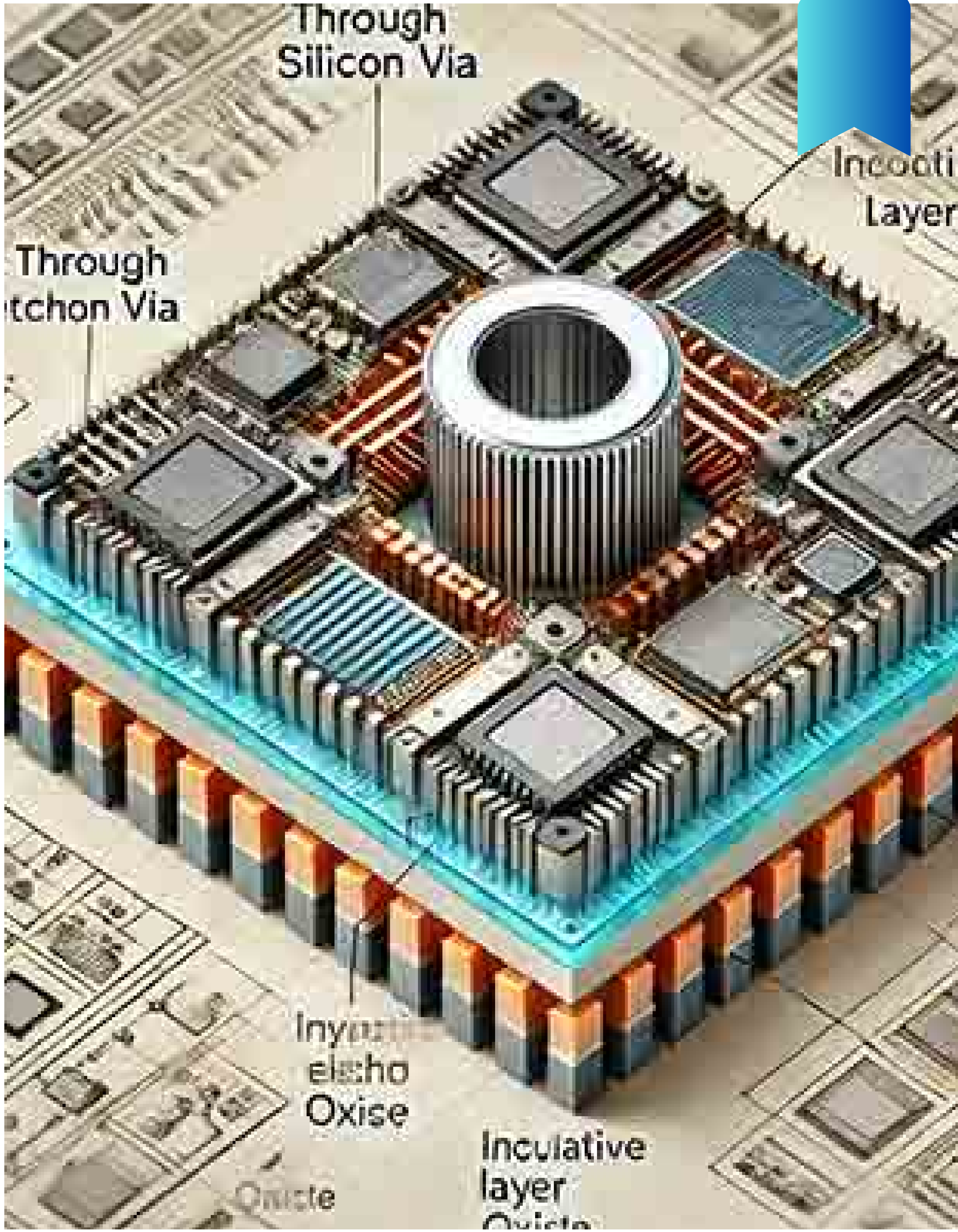
Progress:

29th May to 7th June



By: Kamineni Dhana Sri Keerthi

Via First, Middle & Last

Via First: TSVs are formed before transistor (CMOS) fabrication.

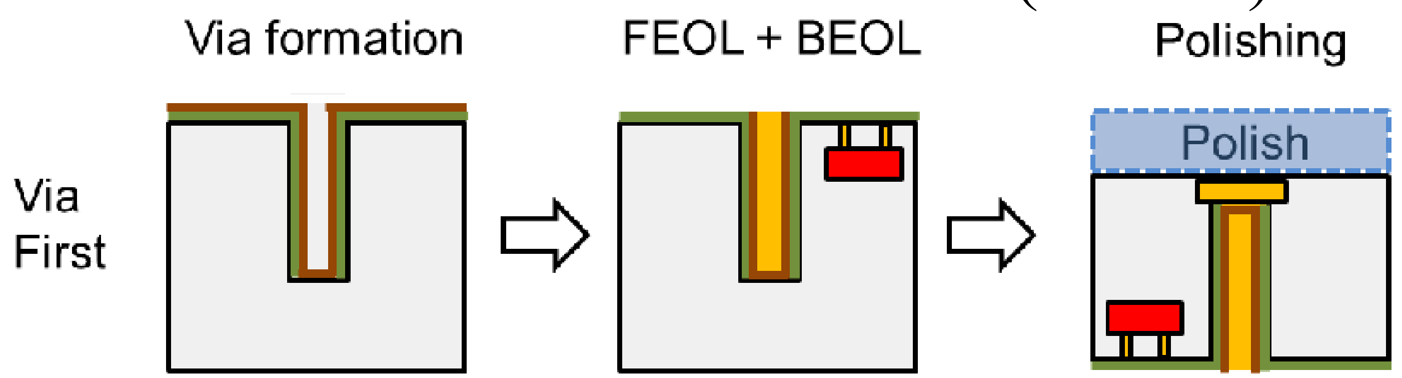


Fig.1.a.

Via Middle: TSVs are formed after transistor fabrication but before metal interconnect layers.

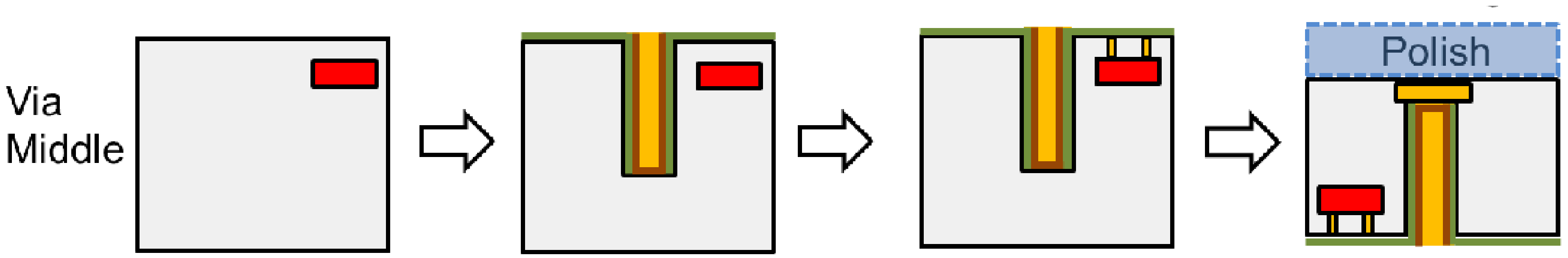


Fig.1.b.

Via Last: TSVs are formed after the complete CMOS fabrication process.

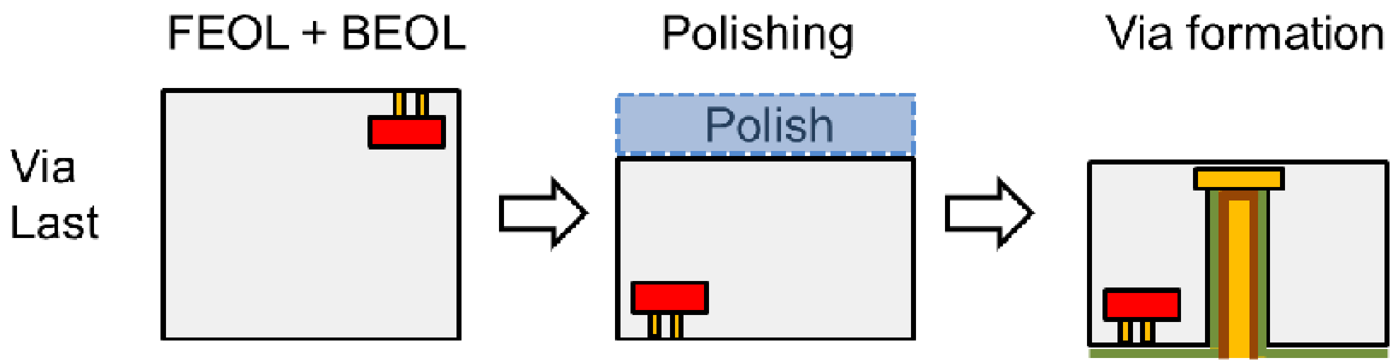


Fig.1.c.

Fig.1 : a.Via First, 1.b. Via Middle, 1.c. Via Last

Why Choose Via Last for Fabrication?

TSV Type	Temperature Exposure	Reason
Via-First	High (~900–1100°C)	TSVs experience all CMOS processing temperatures after fabrication.
Via-Middle	Moderate (~300–500°C)	TSVs are formed after transistor fabrication, reducing thermal stress.
Via-Last	Low (<400°C)	TSVs are fabricated after CMOS processing, minimizing thermal exposure.

- TSV fabrication occurs after CMOS transistor fabrication, so the transistors are not exposed to additional high temperatures.
- Reduces thermal stress and potential damage to devices.

Fabrication Work Flow

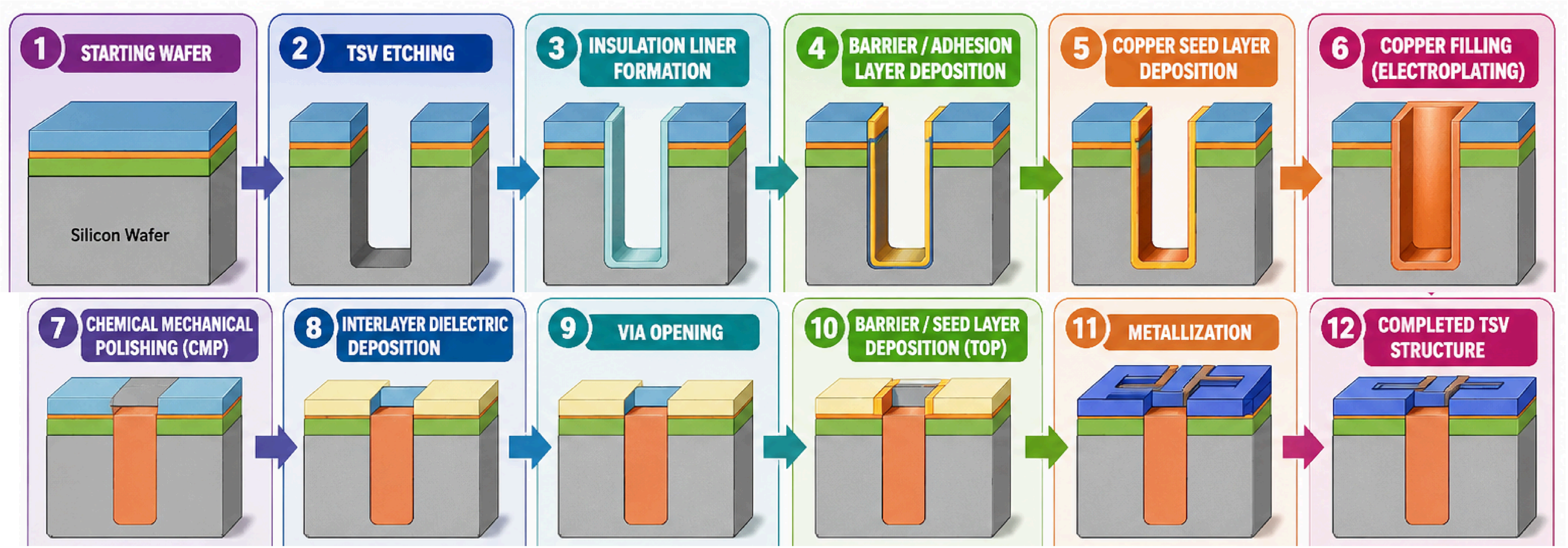


Fig.2. Via Fabrication Process.

Types of Defects

Void Defect: A void defect is defined as a tiny gap that forms inside the copper filled TSV during the filling process.

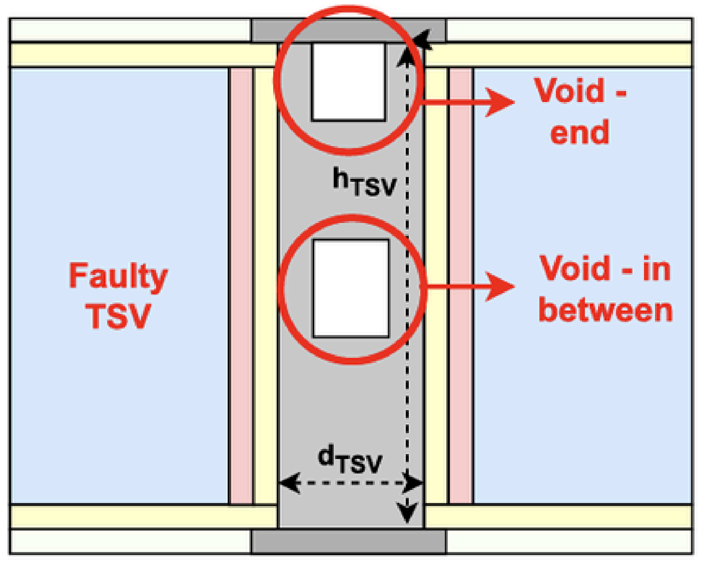
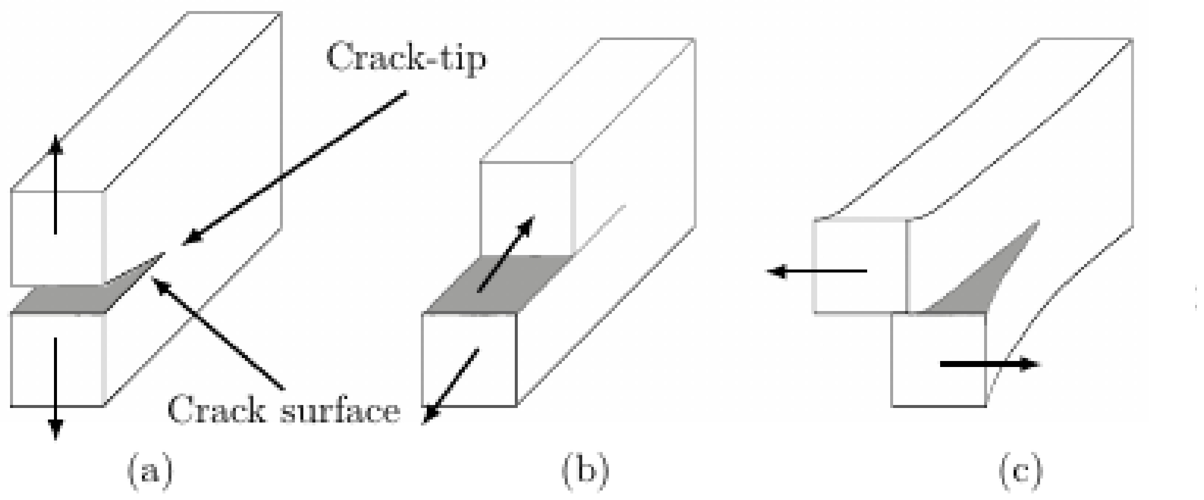


Fig.3.a. Void Defect(3)

Delamination Defect: The Defect that occurs when the layers inside the chip start to crack due to Thermal Stress



Pinhole Defect: A pinhole defect is defined as a tiny hole in the insulator or barrier that causes leakage current to flow from TSV to the Si Substrate.

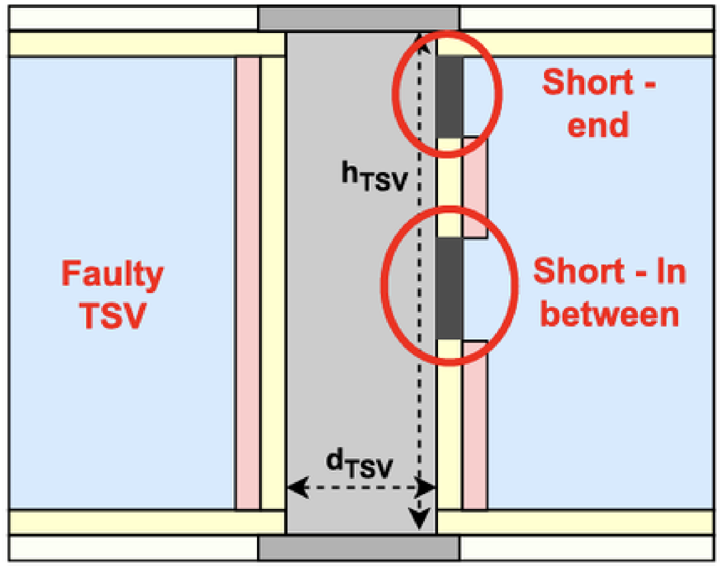


Fig.3.b. PinHole Defect(3)

Types of Defects

Disconnection Failure: Disconnection failures are caused by the impurities between the TSV's and Bumps

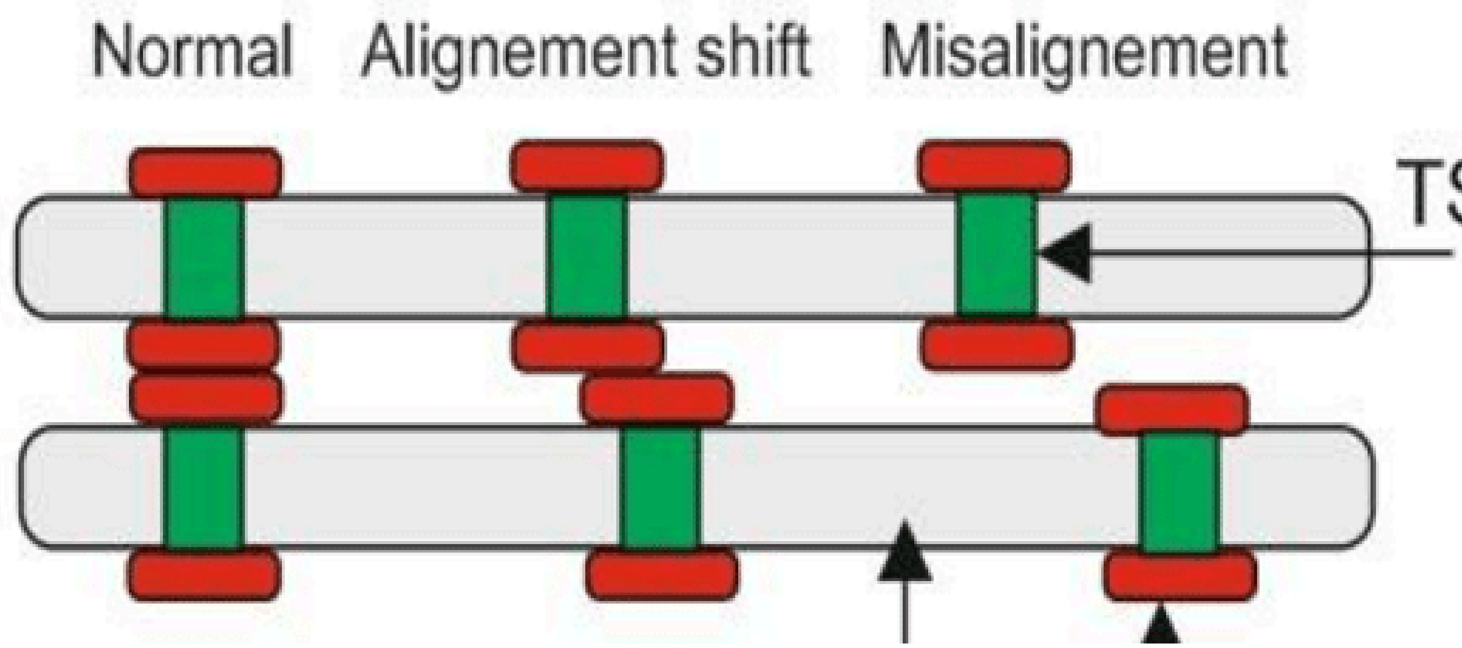


Fig.3.c. Misalignment Defect

Misalignment: A misalignment defect is the displacement of a TSV from its intended position

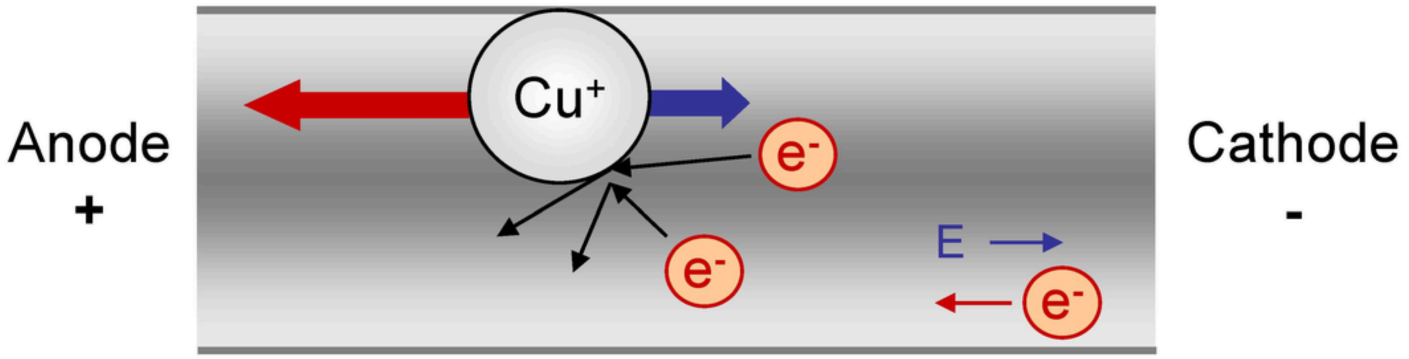


Fig.3.d. ElectroMigration

Electro-Migration: Electromigration is the movement of metal atoms caused by the flow of high electrical current through a thin conductor.

Focusing PinHole Defect

- Pinhole defects are one of the most common dielectric liner defects in TSVs.
- They create an unintended leakage path between the TSV and silicon substrate.
- They increase leakage current and leakage power.
- They degrade signal integrity by affecting transmission characteristics.
- They can increase noise, crosstalk, and propagation delay.
- They reduce the overall reliability of 3D ICs.

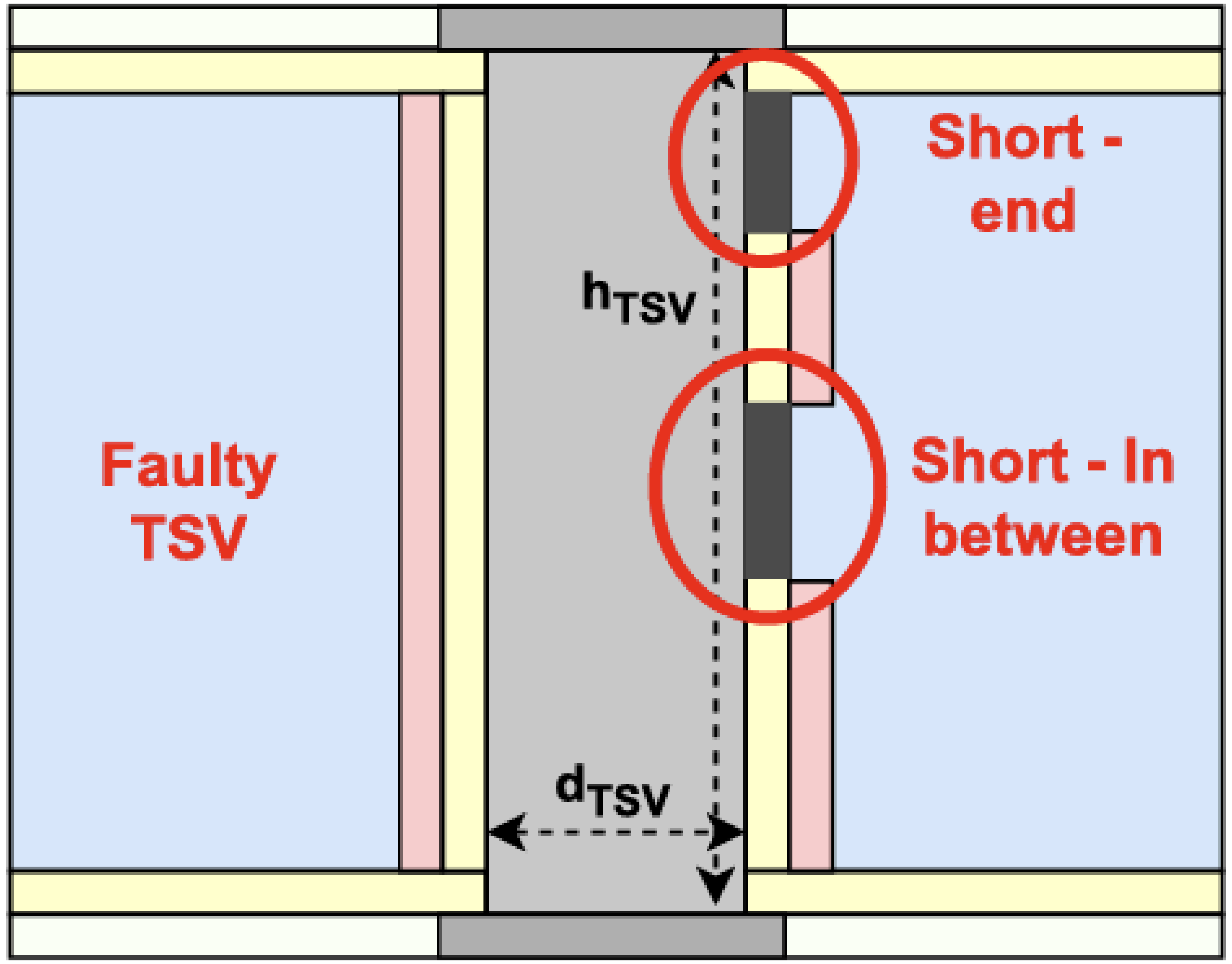


Fig.4. PinHole Defect[3]

Literature Survey

S.No	Paper	Methodology	Results	Observation
01.	High-Frequency Scalable Electrical Model and Analysis of a Through Silicon Via[1]	Developed a scalable equivalent circuit model using RLGC (Resistance, Inductance, Conductance, and Capacitance) parameters to predict TSV electrical behavior.	The frequency dependent loss of a TSV channel, which is capacitive and resistive, is also analyzed in the time domain by eye-diagram measurements.	The work focuses on TSV electrical modeling and behaviour. Further analysis of TSV defects, such as pinhole defects, and their impact on leakage current, leakage power, noise, crosstalk, and delay can be explored.
02.	Reliability Concerns in Through Silicon Via Based 3D Integration: Fabrication to Packaging[2]	Studied TSV reliability challenges from fabrication to packaging	Identified major reliability issues such as thermal stress, electromigration, cracking, voids, and TSV failures.	The electrical impact of individual TSV defects can be further investigated
03.	A Novel TSV Model With Fault Characterization for High-Frequency Transmission in 3D ICs[3]	Developed a comprehensive equivalent circuit model for fault-free and faulty TSVs	Demonstrated how TSV faults affect transmission performance and signal integrity at high frequencies.	Additional performance metrics such as leakage current, leakage power, noise, crosstalk, and delay can be further investigated.
04.	Modelling and simulation of TSV considering void and leakage defects.[4]	This study establishes a unified equivalent circuit model that includes leakage defect TSV, void defect TSV, and defect-free TSV	Demonstrated that void and leakage defects degrade TSV electrical performance and affect signal transmission characteristics.	The study focuses on void and leakage defects. Detailed analysis of pinhole defects, including leakage current, leakage power, crosstalk, noise, and delay, can be further explored.

Why Choose Cylindrical TSV?

Easier Fabrication:

- TSVs are formed using DRIE (Deep Reactive Ion Etching).
- DRIE naturally produces nearly circular holes.

Uniform Electrical Field Distribution:

- Electric field lines spread symmetrically around the TSV.
- Reduced field concentration at edges.

Less Concentration:

- Since a cylinder has no sharp corners, the field lines doesn’t get concertration.
- So it reduces Electrical Stress.

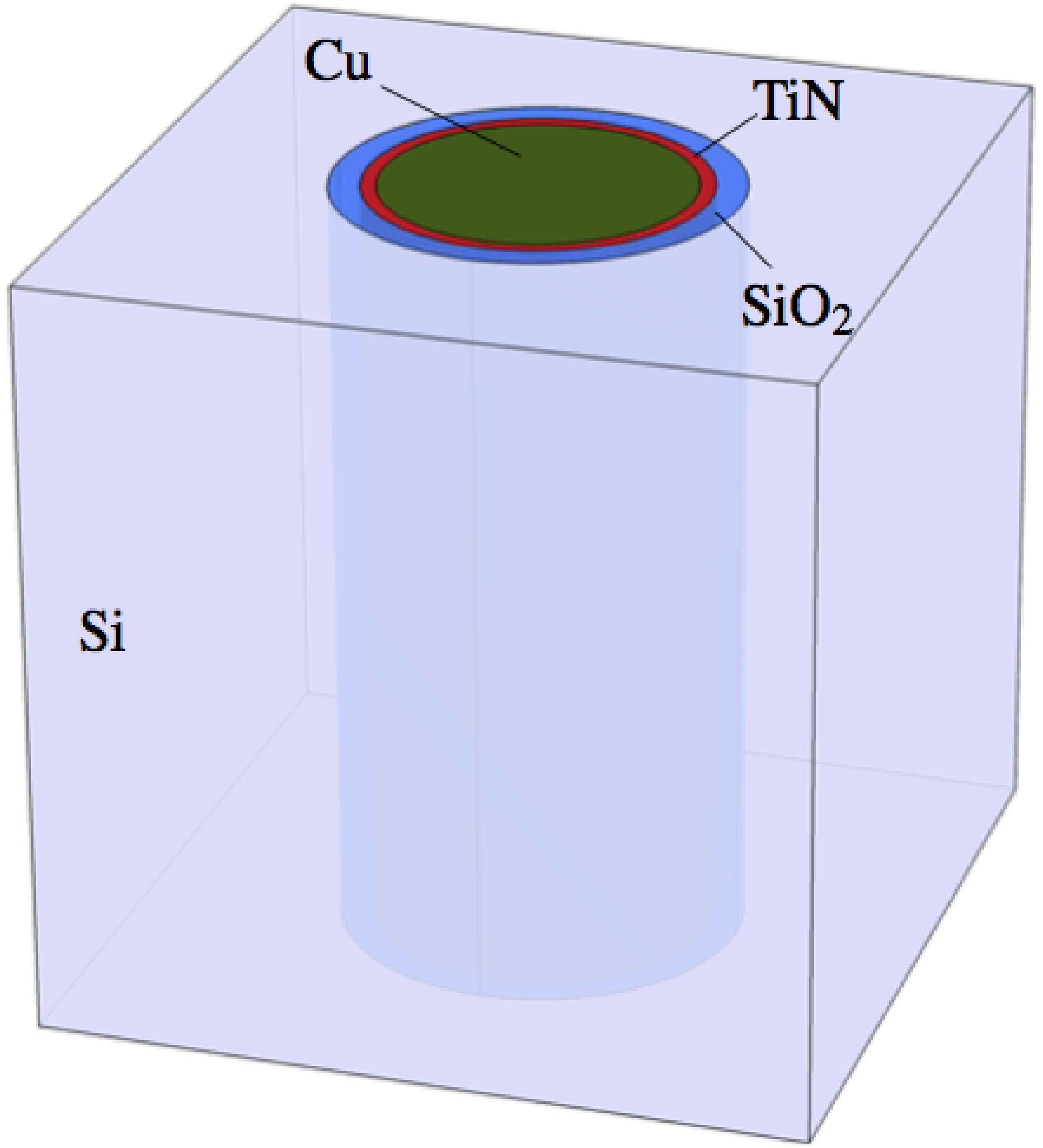


Fig.1 Cylindrical Through Silicon via

Types of Networks

Pi Network:

- Consists of one series element and two shunt element.
- The Shunt Element gets divided into half.

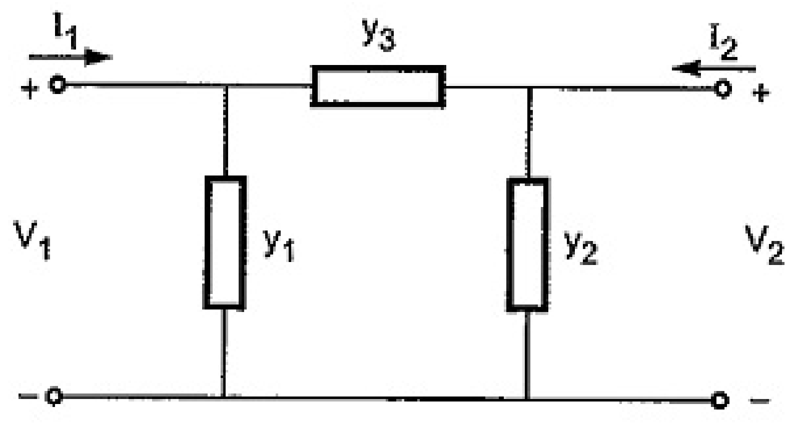


Fig.2.a. PI Network

T Network:

- Consists of two series element and one shunt element.
- The Series Element gets divided into half.

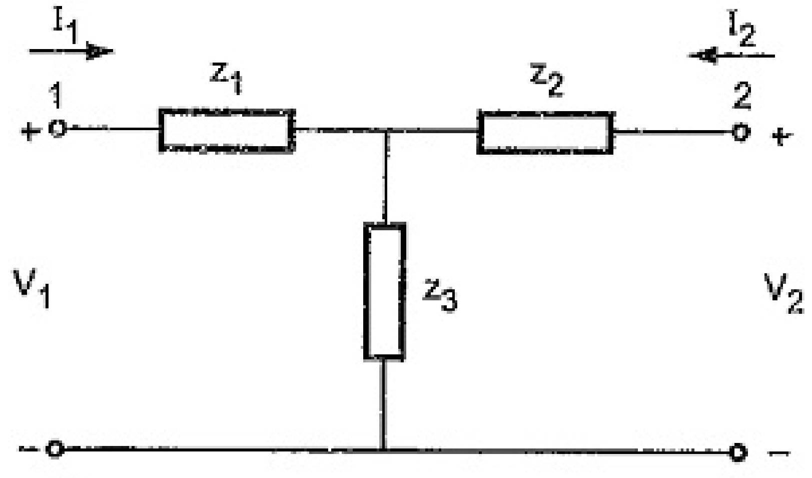


Fig.2.b. T Network

L Network:

- Consists of one series element and one shunt element.
- Shape resembles the letter L.

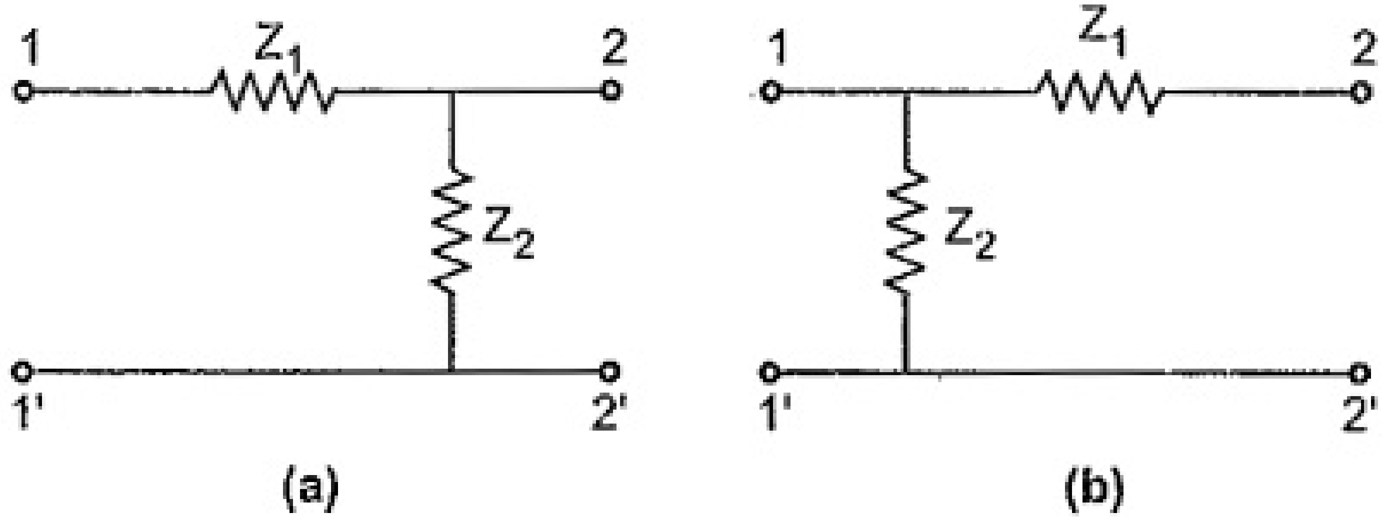


Fig.2.c. T Network

Types of Electrical Models

Lumped Model:

- Electrical components are concentrated at discrete points.
- Voltage and current are considered uniform throughout each component
- If the circuit size is much smaller than the wavelength
- Example: R,L,C.

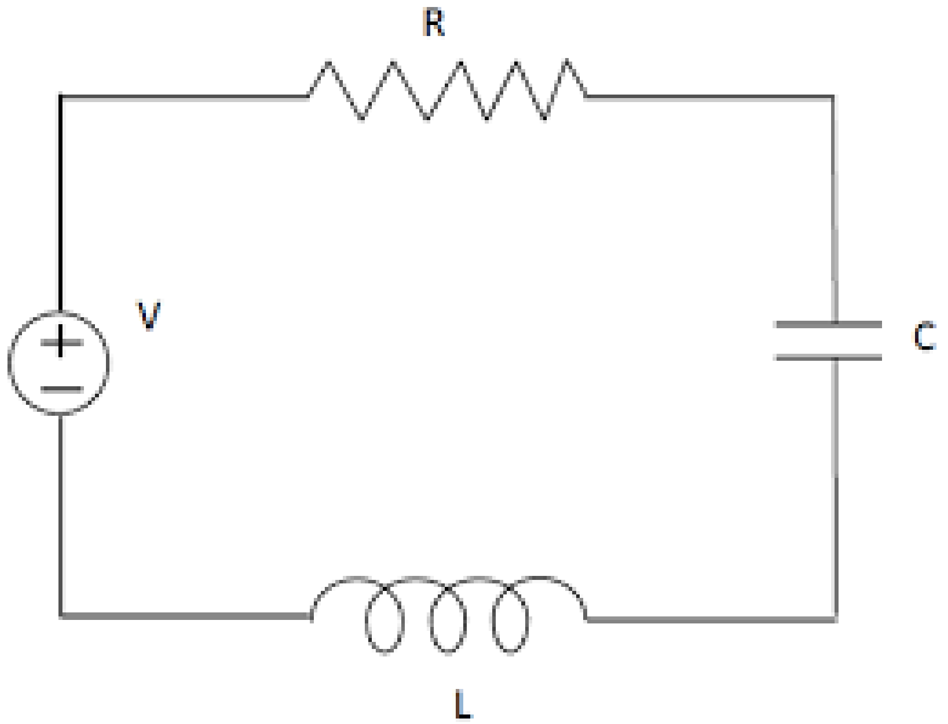


Fig.3.a. Lumped Model

Distributed Model:

- Electrical parameters are distributed continuously along the structure.
- Voltage and current vary with position and time.
- Required when physical dimensions are comparable to the signal wavelength.
- Represented using distributed resistance, inductance, capacitance, and conductance per unit length
- Example: Transmission line

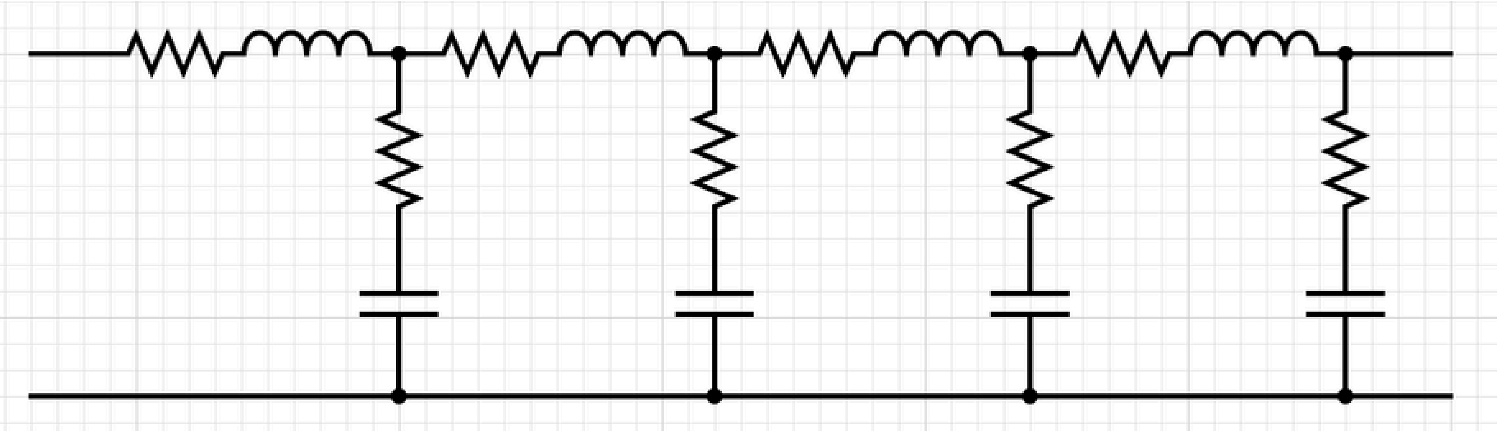


Fig.3.b. Distributed Model

Defect Free TSV Electrical Modelling

RLGC Equations

Circuit Diagram

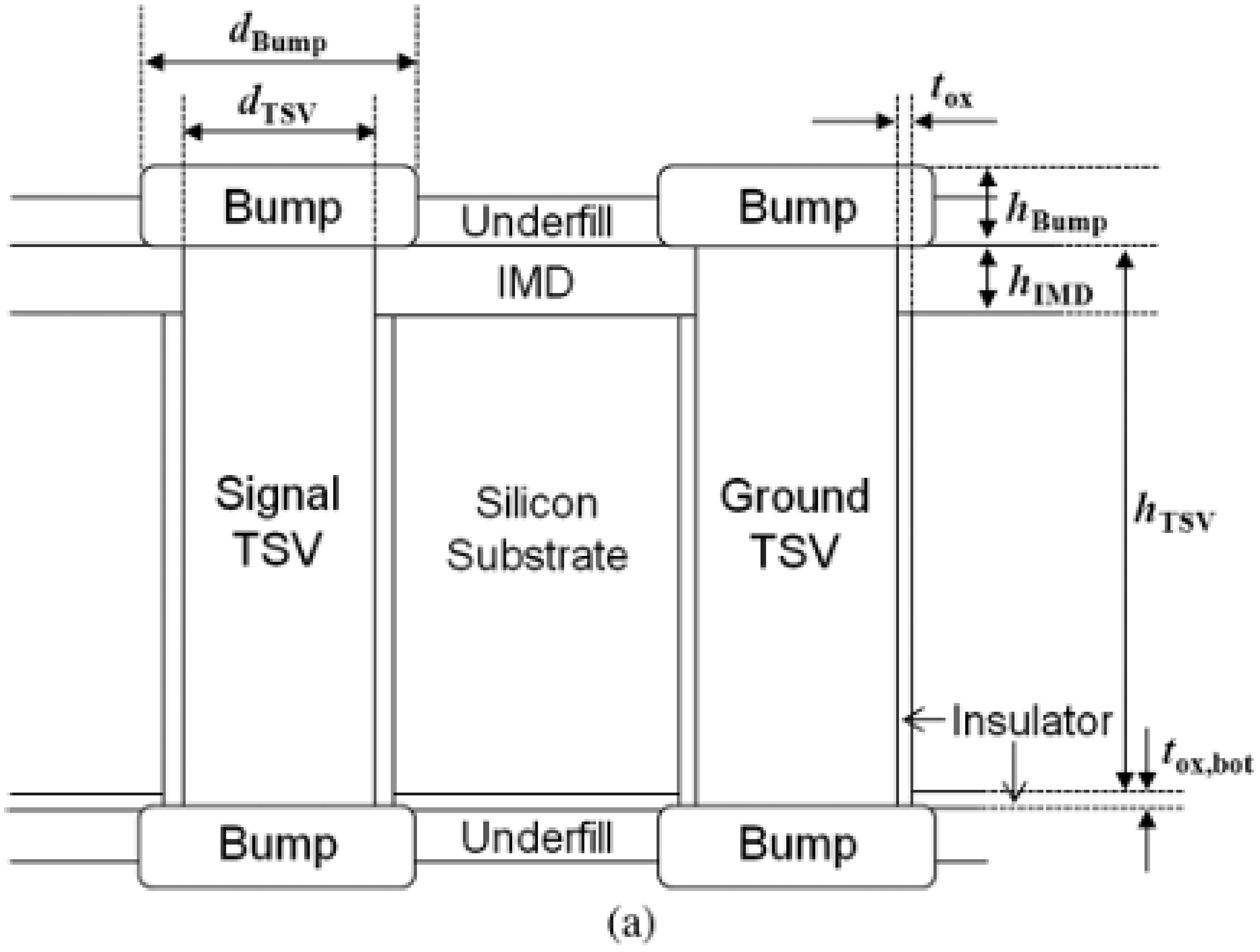


Fig.4 (a) Structure of a signal TSV and a ground TSV with bumps with the via-last process and their structural parameters[(1)]

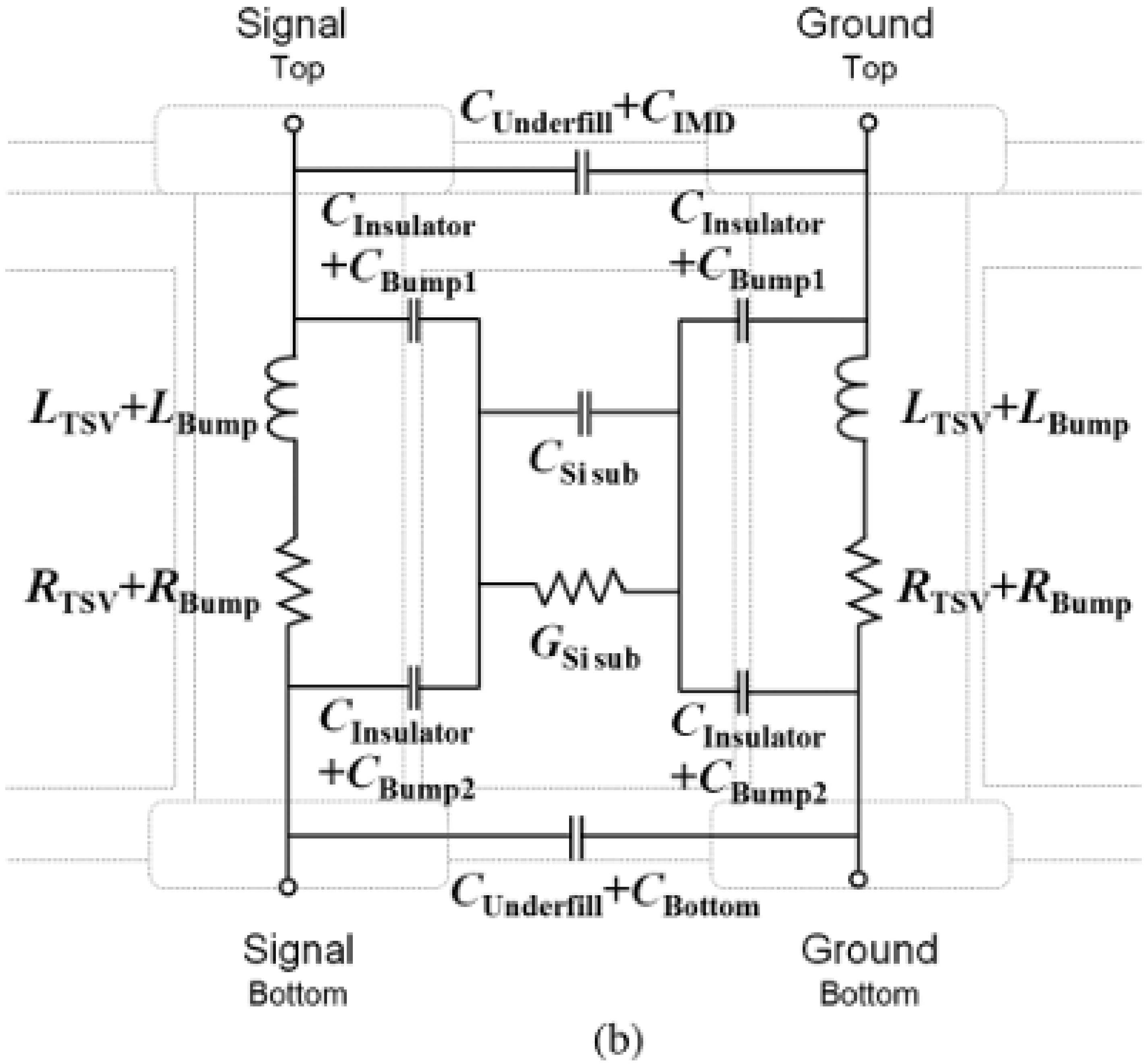


Fig.4 (b) The scalable electrical model of Fig.4(a) with labelled components[(1)]

Types of Capacitances

- Insulator Capacitance
- IMD Capacitance
- Bump01 Capacitance
- Bottom Capacitance
- Bump02 Capacitance
- Substrate Capacitance
- Underfill Capacitance

Insulator Capacitance

$$C_{\text{Insulator}} = \frac{1}{2}\left\{2\pi \times \varepsilon_0 \varepsilon_{r,\text{Insulator}} \times \frac{h_{\text{TSV}} - h_{\text{IMD}}}{\ln\left(\frac{d_{\text{TSV}}/2 + t_{\text{ox}}}{d_{\text{TSV}}/2}\right)}\right\}[F]. \quad (1)$$

- Insulator Capacitance is a Cylindrical Capacitance
- TSV - INSULATOR - Silicon Substrate
- The General Formula for a Cylindrical Capacitance is

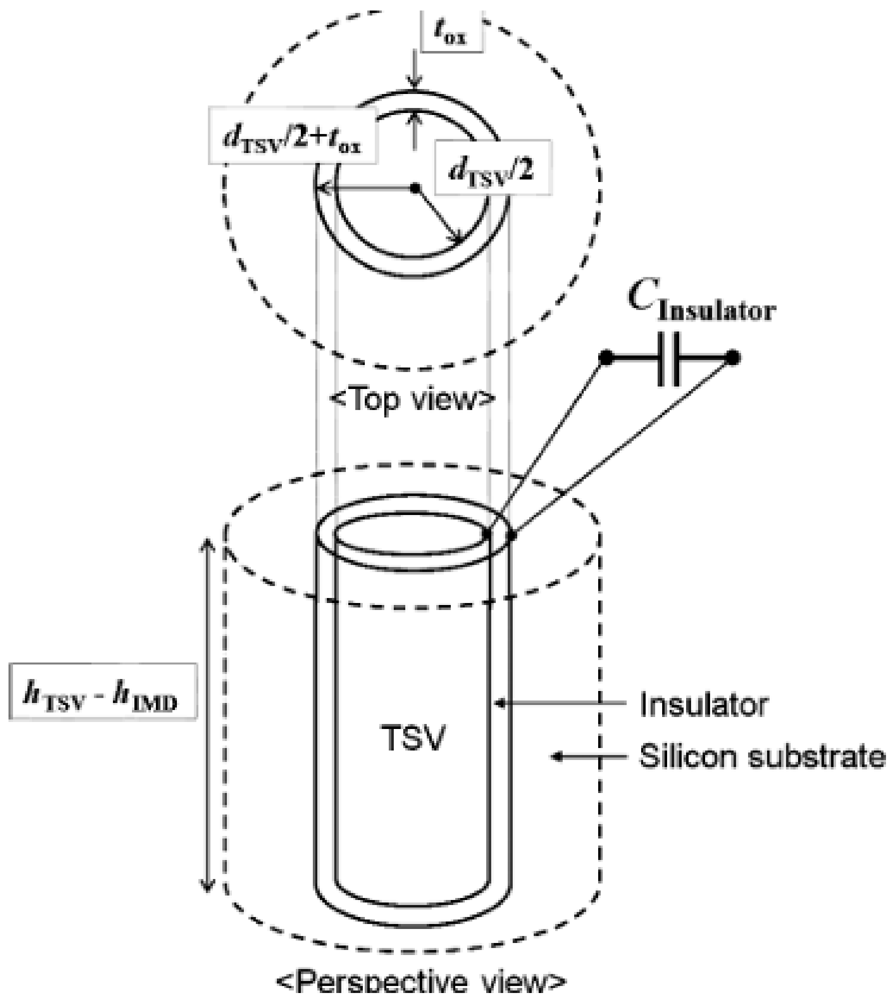


Fig.5.a. Top view and a perspective view of a TSV[1]

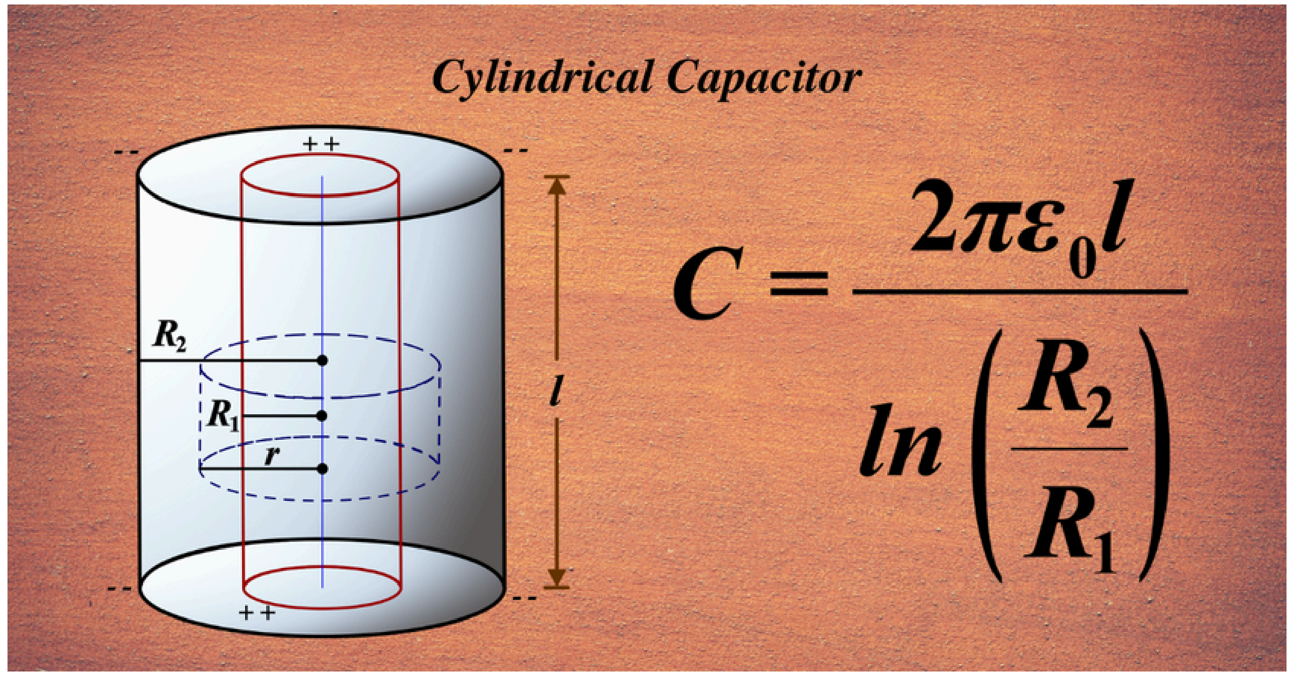


Fig.5.b. Cylindrical capacitor

Bump01 Capacitance

$$C_{\text{Bump1}} = \varepsilon_0 \varepsilon_{r,\text{IMD}} \times \frac{\pi \times \left\{(d_{\text{Bump}}/2)^2 - (d_{\text{TSV}}/2 + t_{\text{ox}})^2\right\}}{h_{\text{IMD}}} [F]$$

- Bump01 Capacitance is a Parallel Capacitance
- Bump - IMD - Silicon Substrate
- The General Formula for a Cylindrical Capacitance is

$$C = \frac{\varepsilon_0 A}{d}$$

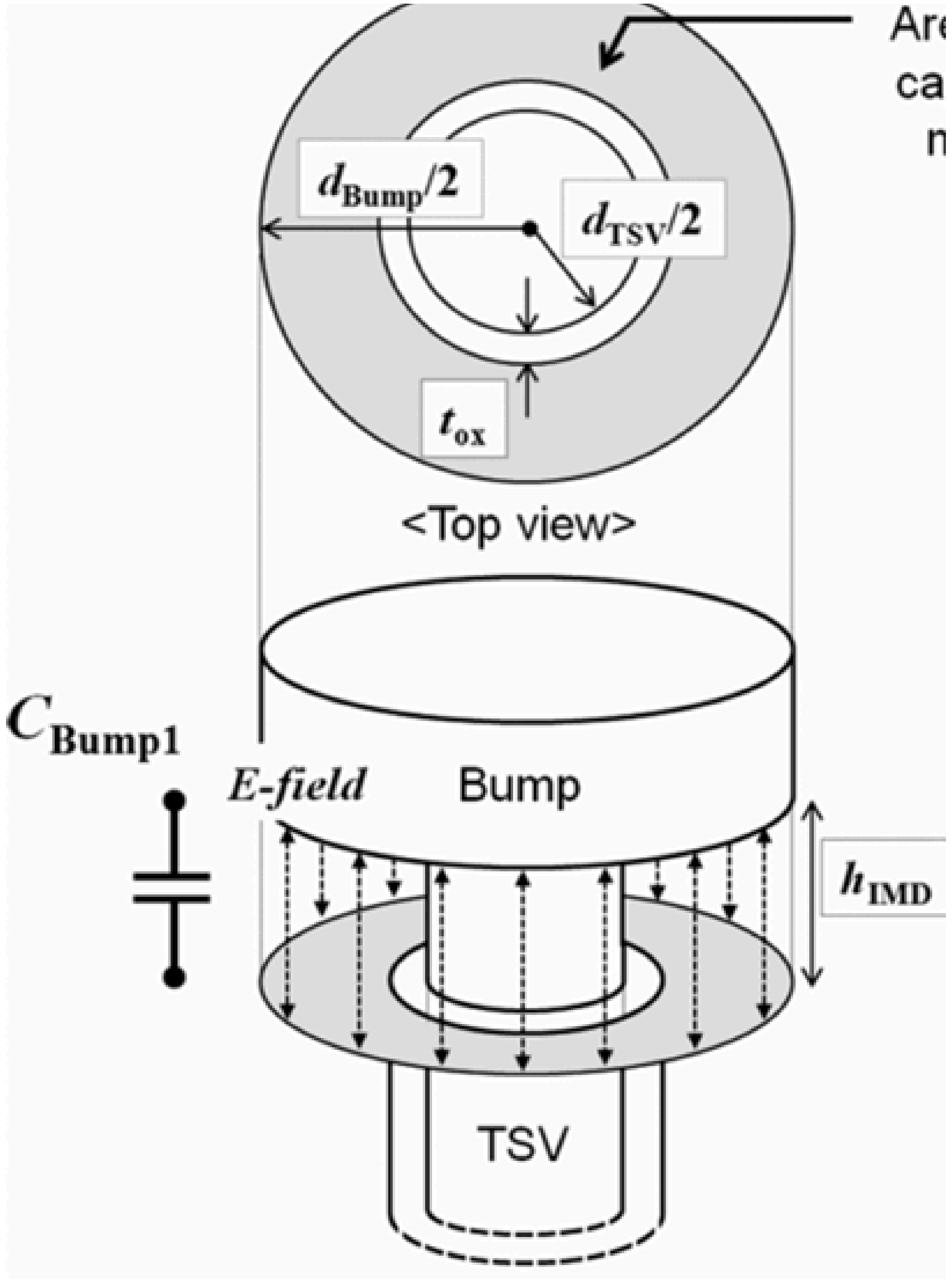


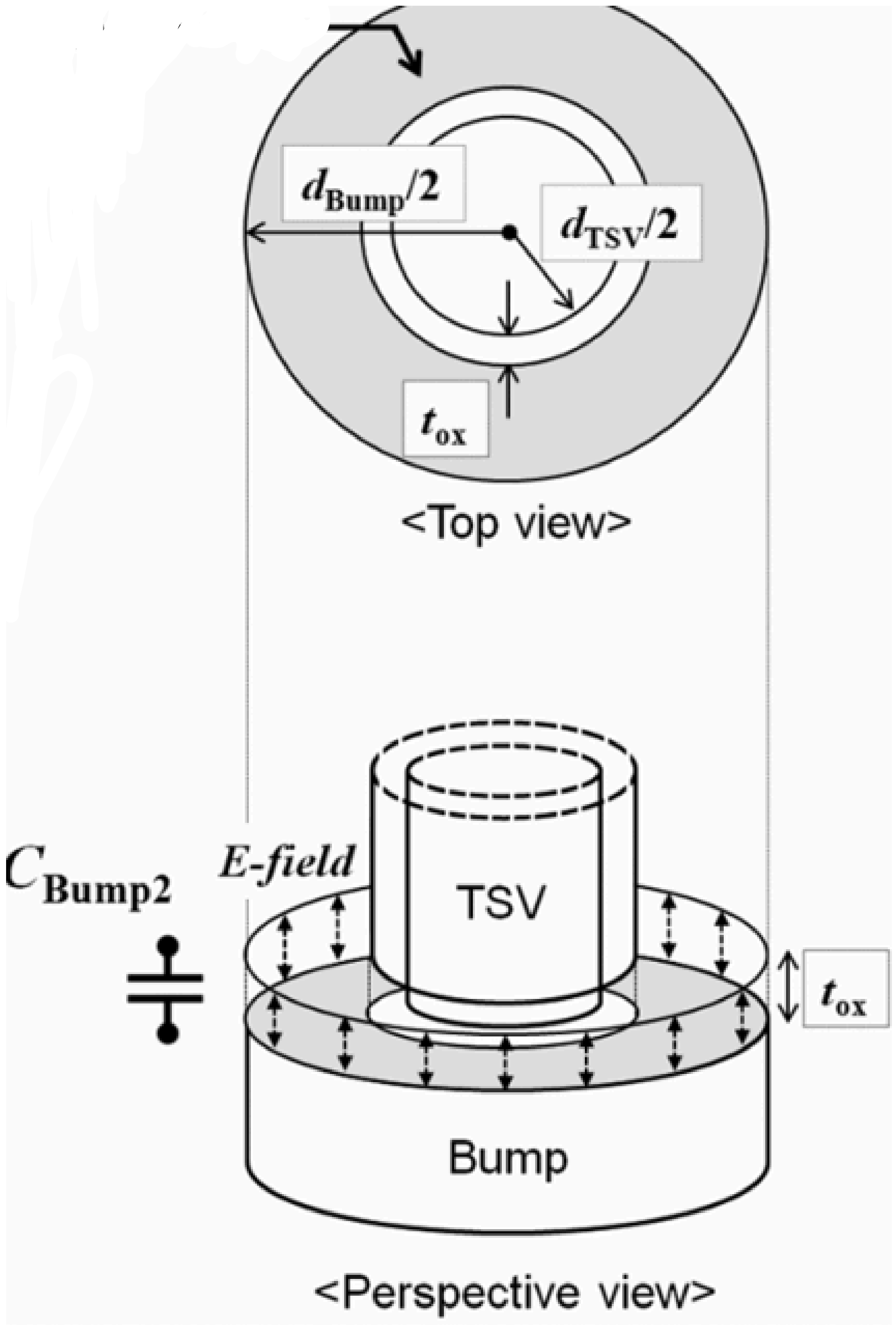
Fig.6.a. A top view and a perspective view of the upper part of the TSV (1)

Bump02 Capacitance

$$C_{\text{Bump2}} = \varepsilon_0 \varepsilon_{r,\text{ox}} \times \frac{\pi \times \{(d_{\text{Bump}}/2)^2 - (d_{\text{TSV}}/2 + t_{\text{ox,bot}})^2\}}{t_{\text{ox,bot}}} [F].$$

- Bump02 Capacitance is a Parallel Capacitance
- Bump - Bottom - Silicon Substrate
- The General Formula for a Cylindrical Capacitance is

$$C = \frac{\varepsilon_0 A}{d}$$



(b)

Fig.6.b. A top view and a perspective view of the bottom part of the TSV (1)

Underfill Capacitance

$$C_{\text{Underfill}} = \frac{\pi \times \varepsilon_0 \varepsilon_{r,\text{Underfill}}}{\cosh^{-1}\left(\frac{p_{\text{TSV}}}{d_{\text{Bump}}}\right)} \times h_{\text{Bump}} [F]$$

- Underfill Capacitance is the capacitance between two parallel capacitances
- Bump - Underfill - Bump
- The General Formula for a Parallel Cylinders Capacitance is

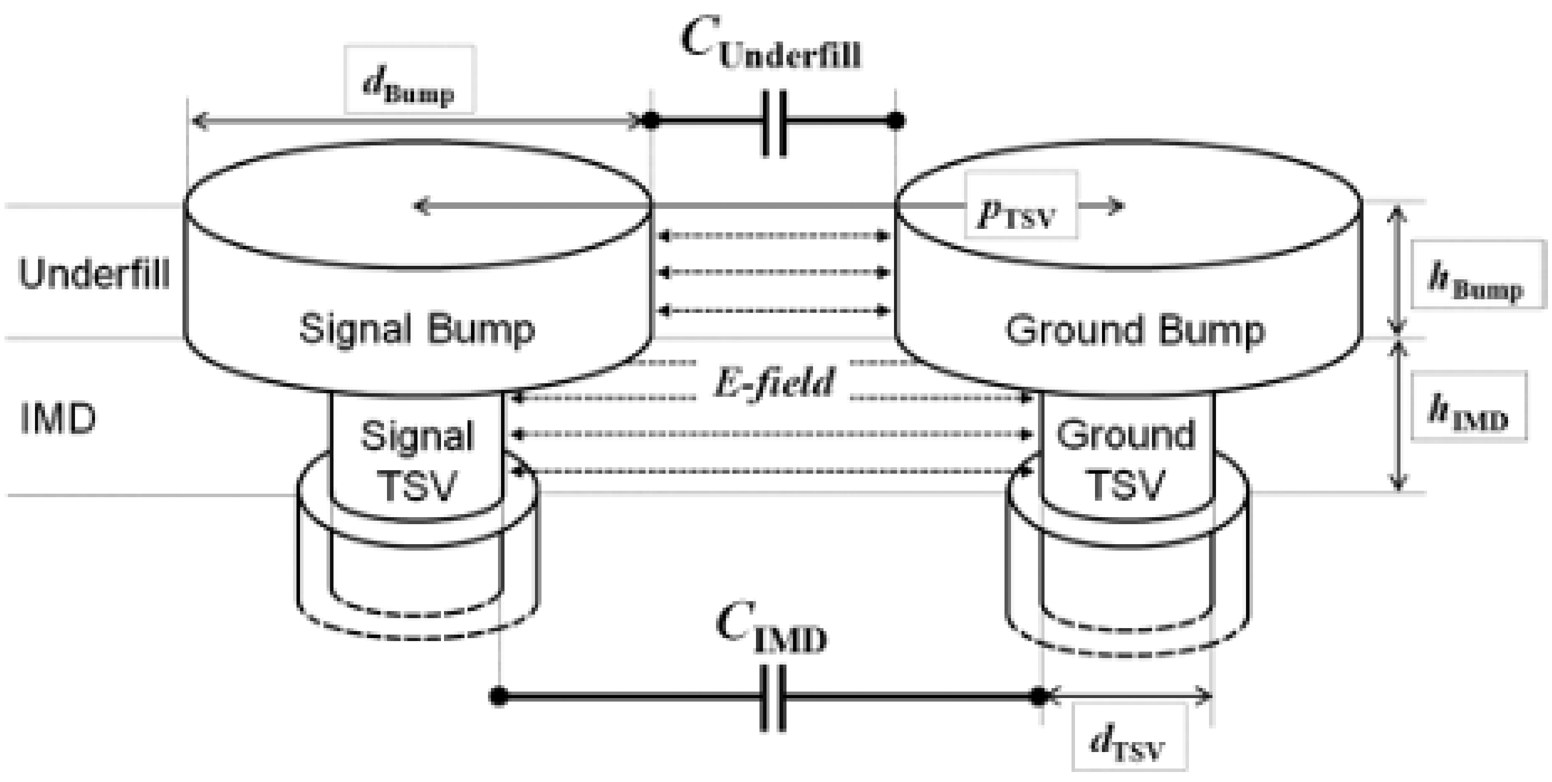


Fig.7.a. Underfill Capacitance between two bumps[(1)]

$$C = \frac{\pi \varepsilon L}{\cosh^{-1}\left(\frac{d}{2a}\right)}$$

Fig.7.b. Formula for a Parallel Plate capacitor

IMD ; Bottom ; Si Substrate Capacitance

Similar to the underfill capacitance IMD, Bottom, Si Substrate capacitances are the capacitance between two parallel cylinders.

IMD Capacitance: [TSV - IMD - TSV]

$$C_{\mathrm{IMD}} = \frac{\pi \times \varepsilon_0 \varepsilon_{r,\mathrm{IMD}}}{\cosh^{-1}\left(\frac{p_{\mathrm{TSV}}}{d_{\mathrm{TSV}}}\right)} \times h_{\mathrm{IMD}}[F]$$

Bottom Capacitance: [TSV - Bottom -TSV]

$$C_{\mathrm{Bottom}} = \frac{\pi \times \varepsilon_0 \varepsilon_{r,\mathrm{ox,bot}}}{\cosh^{-1}\left(\frac{p_{\mathrm{TSV}}}{d_{\mathrm{TSV}}}\right)} \times t_{\mathrm{ox,bot}}[F].$$

Si Sub Capacitance: [TSV - Si Substrate - TSV]

$$C_{Si\ \mathrm{sub}} = \frac{\pi \times \varepsilon_0 \varepsilon_{r,Si}}{\cosh^{-1}\left(\frac{p_{\mathrm{TSV}}}{d_{\mathrm{TSV}}}\right)} \times (h_{\mathrm{TSV}} - h_{\mathrm{IMD}})[F].$$

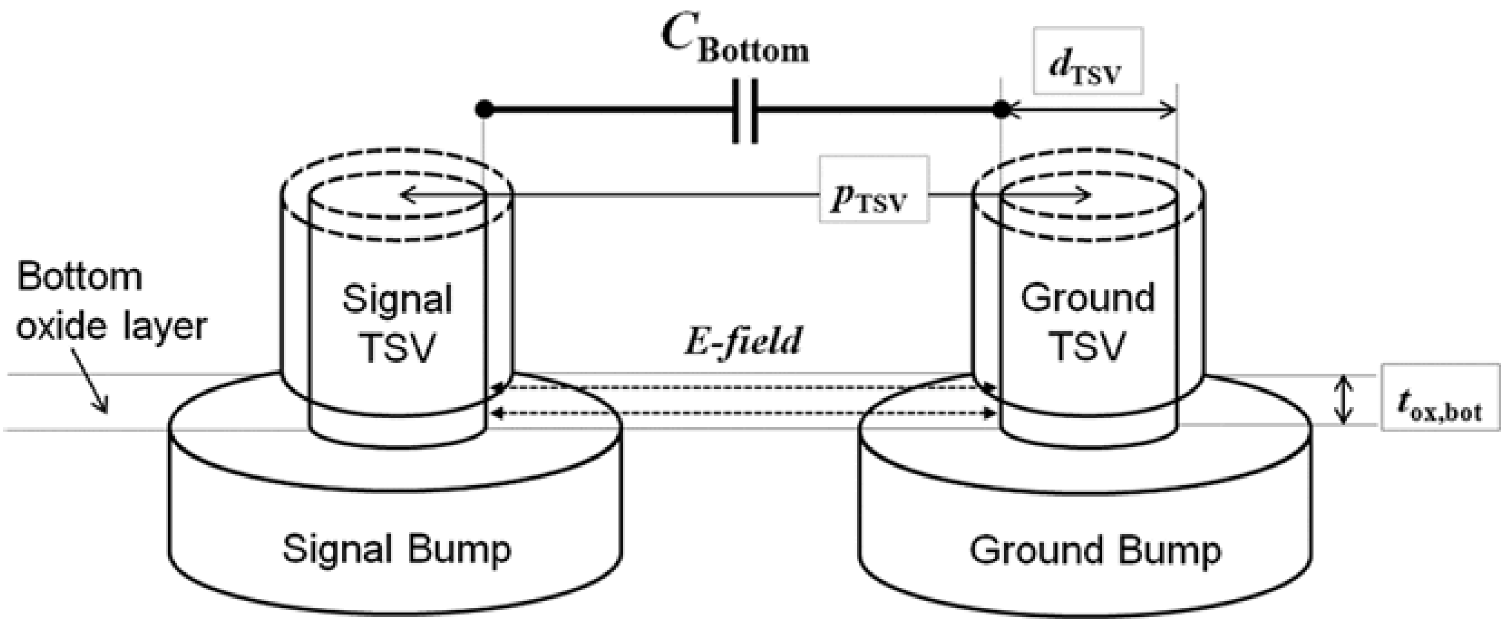


Fig.8.a. Bottom capacitance between GTSV and STSV(1)

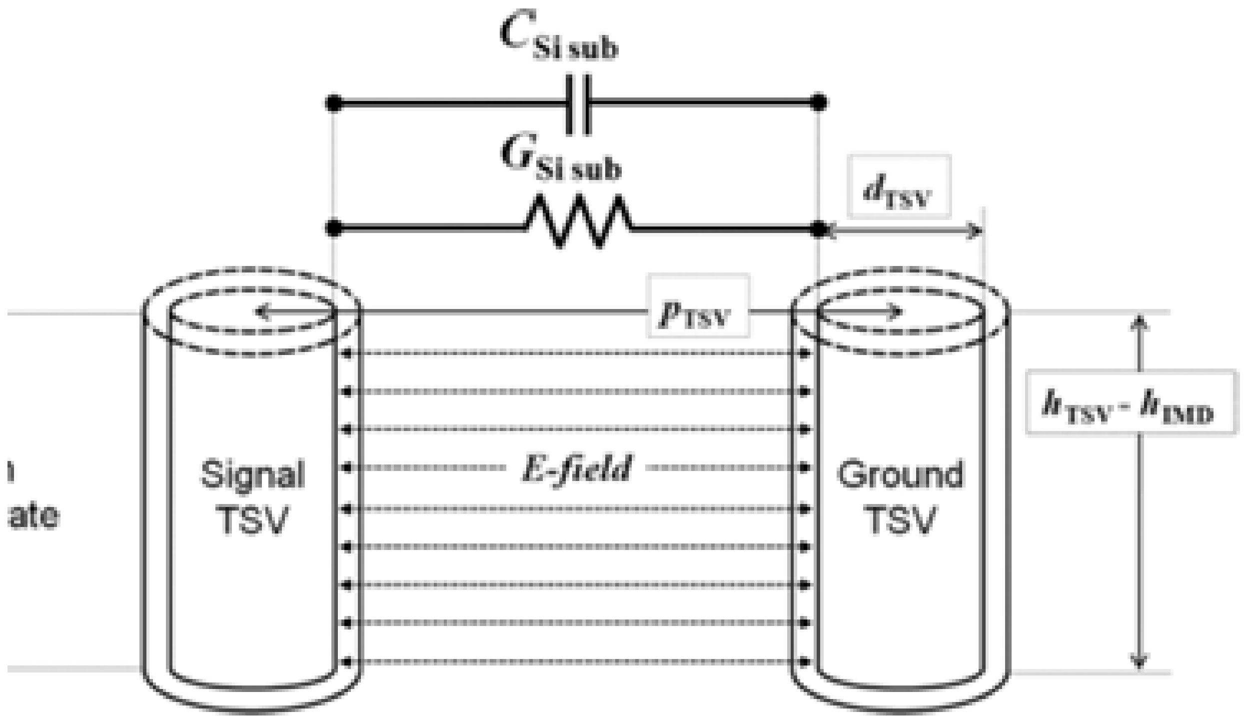


Fig.8.b. si substrate capacitance between GTSV and STSV(1)

Inductance of TSV

Self Indcutance:

- General formula for self inductance of a cylinder

$$L = \left(\frac{\mu_o \mu_r l}{2\pi}\right) \ln\left(\frac{b}{a}\right) [H]$$

- The Self Inductance of Single TSV:

$$L_{TSV} = \frac{1}{2}\left(\frac{\mu_o \mu_{r,TSV} h_{TSV}}{2\pi}\right) \ln\left(\frac{p_{TSV}}{r_{TSV}}\right) [H]$$

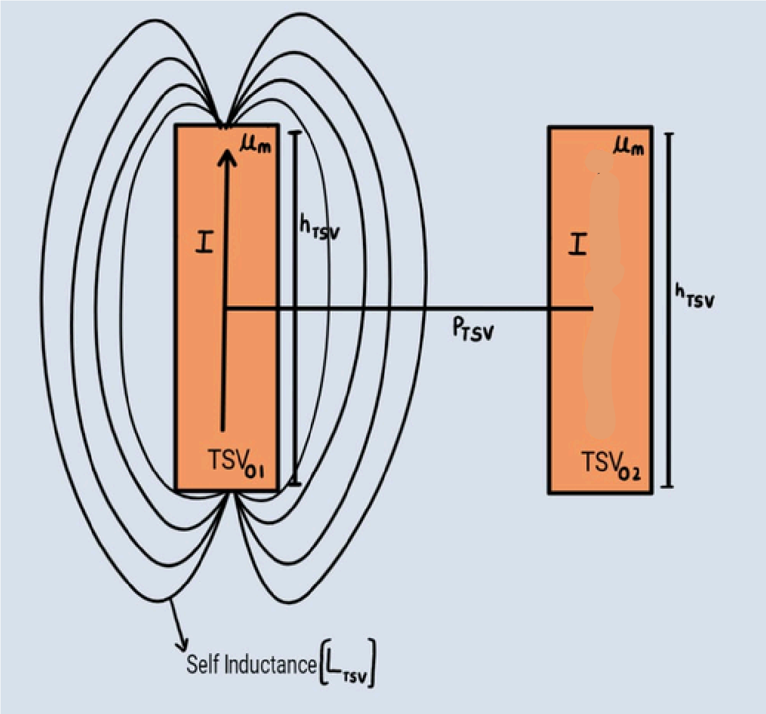


Fig.9.a Self Inductance of Single TSV

Mutual Indcutance:

- General formula for mutual inductance between two cylinders: $L_M = \left(\frac{\mu_o \mu_r l}{2\pi}\right)\left(\ln\left(\frac{2l}{p}\right) - 1\right) [H]$

- The Mutual Inductance of two TSV's:

$$L_M = \left(\frac{\mu_o \mu_{r,TSV}}{2\pi}\right)\left(\ln\left(\frac{2l_{TSV}}{p_{TSV}}\right) - 1\right) [H]$$

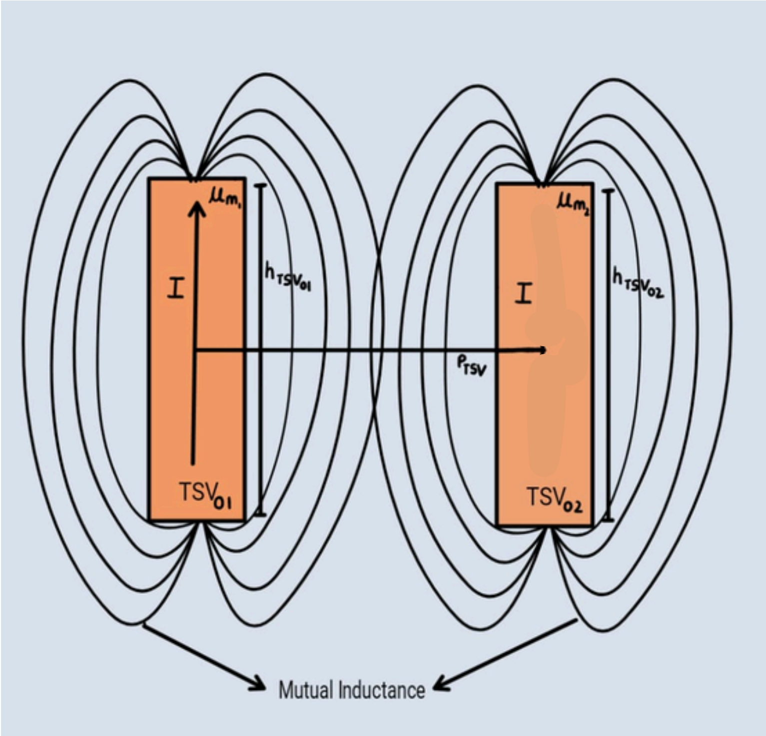


Fig.9.b. Mutual Inductance of two TSV

Resistance of TSV

$$R_{\mathrm{TSV}} = \sqrt{(R_{\mathrm{dc,TSV}})^2 + (R_{\mathrm{ac,TSV}})^2}[\Omega]$$

Where -

$$R_{\mathrm{dc,TSV}} = \rho_{\mathrm{TSV}} \times \frac{h_{\mathrm{TSV}}}{\pi \times \left(\frac{d_{\mathrm{TSV}}}{2}\right)^2}[\Omega]$$

&

$$R_{\mathrm{ac,TSV}} = k_p \left(\rho_{\mathrm{TSV}} \times \frac{h_{\mathrm{TSV}}}{2\pi \times \frac{d_{\mathrm{TSV}}}{2} \times \delta_{\mathrm{skin\ \ depth,TSV}} - \pi\delta^2_{\mathrm{skin\ \ depth,TSV}}} \right)[\Omega]$$

Where -

$$\delta_{\mathrm{skindepth,TSV}} = \frac{1}{\sqrt{\pi f \mu_{\mathrm{TSV}} \sigma_{\mathrm{TSV}}}}[\mathrm{m}]$$

Conductance of TSV

W.K.T $C = \varepsilon K$ & $G = \sigma K$ Where - K is the geometry factor

$$\frac{C_{Si\ \mathrm{sub}}}{G_{Si\ \mathrm{sub}}} = \frac{\varepsilon_{Si}}{\sigma_{Si}}$$

$$G_{Si\ \mathrm{sub}} = \frac{\pi \times \sigma_{Si}}{\cosh^{-1}\left(\frac{p_{\mathrm{TSV}}}{d_{\mathrm{TSV}}}\right)} \times (h_{\mathrm{TSV}} - h_{\mathrm{IMD}})[S].$$

Defected TSV Electrical Modelling

1. Pinhole Defect is Modelled in terms of RL
2. Pinhole Defect is Modelled in terms of C_{Leak} and G_{Leak}

Pinhole Defect in Terms of RL

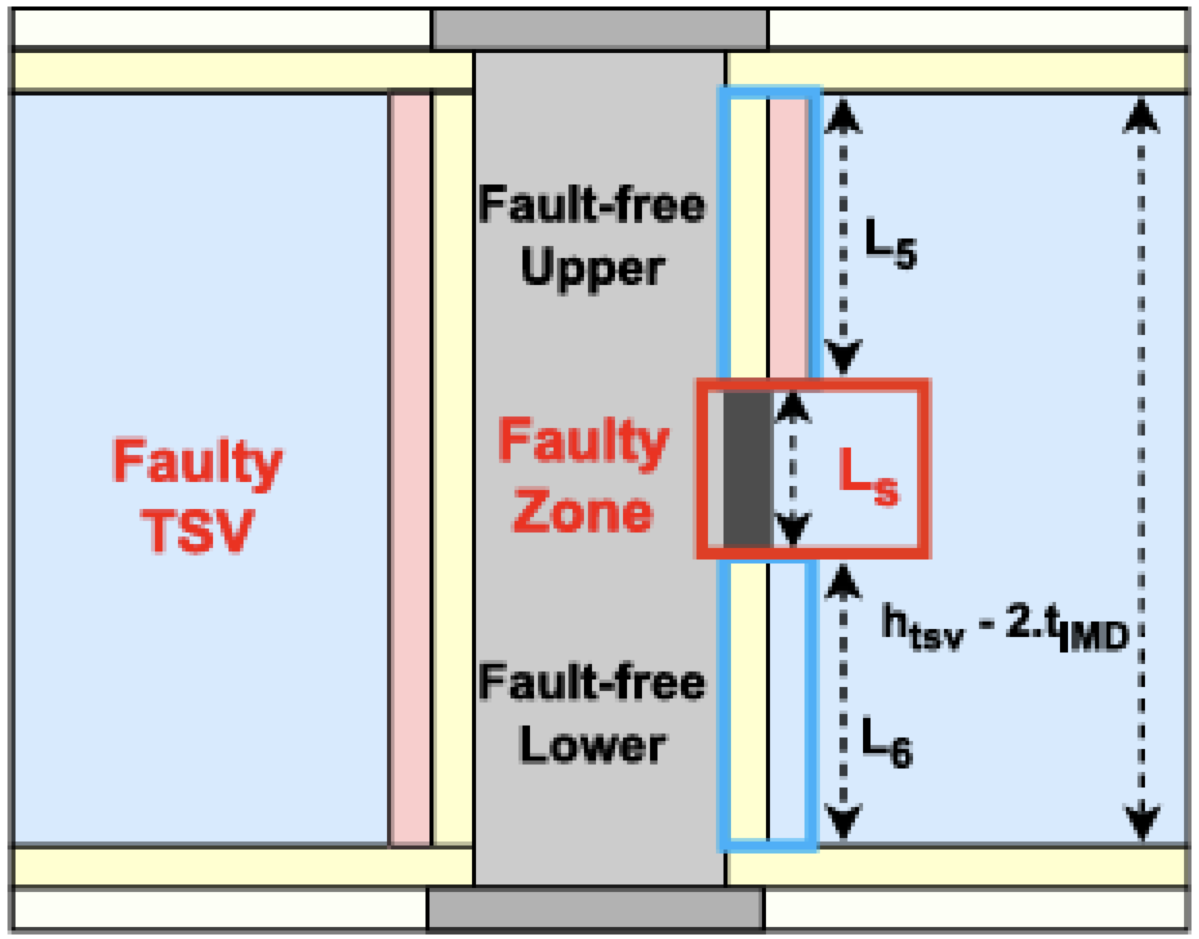


Fig.10.a. Schematic of TSV with short fault.(3)

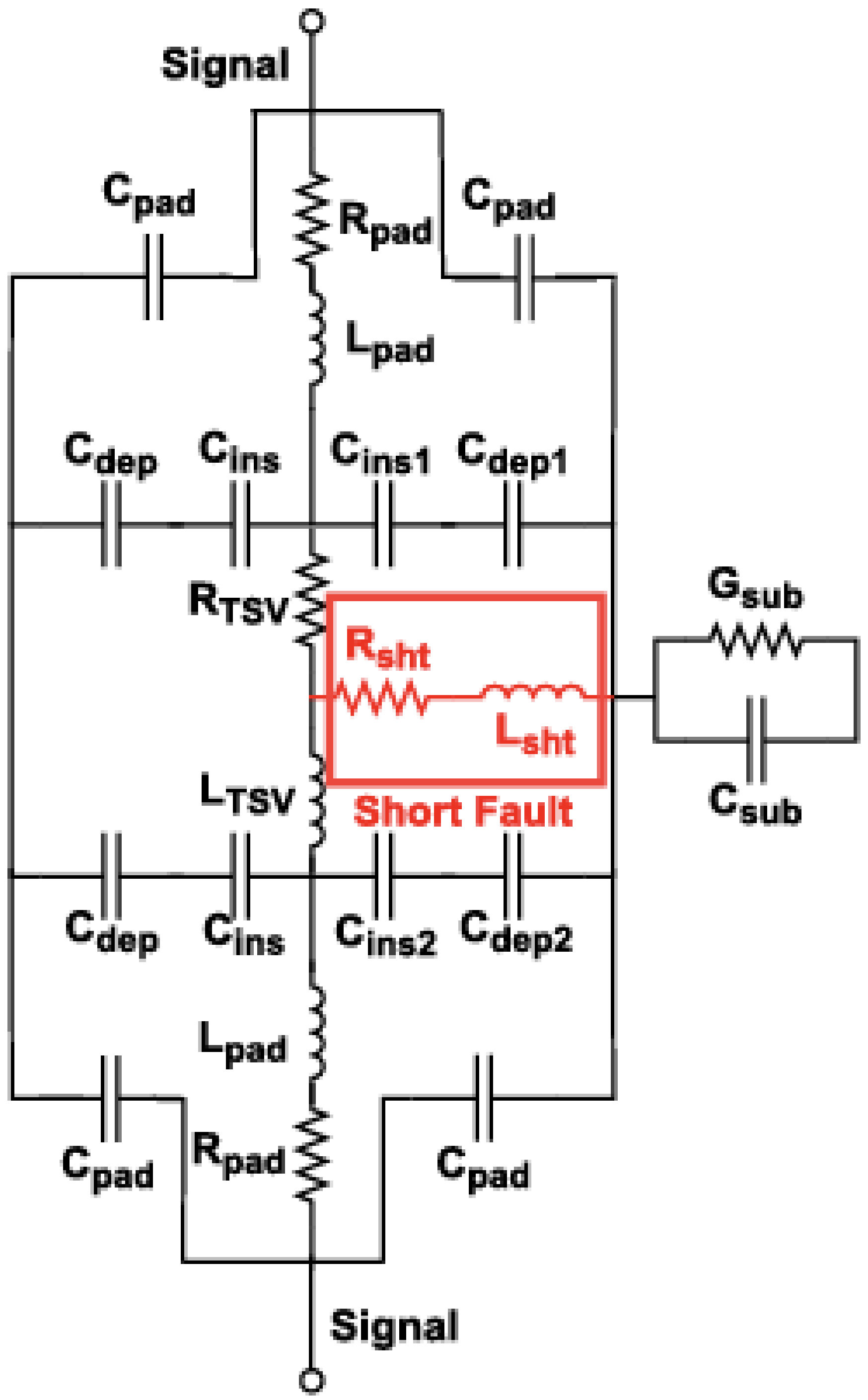


Fig.10.b. Equivalent circuit of TSV with short fault.(3)

R,L Equations

- $$R_{\mathrm{sht}} = \frac{\rho_T \cdot t_{\mathrm{ins}}}{L_s^2}$$

- $$L_{\mathrm{sht}} = \frac{\mu \cdot \mu_{\mathrm{tsv}} \cdot t_{\mathrm{ins}}}{2\pi} \cdot \left(\operatorname{arccosh}\left(\frac{t_{\mathrm{ins}}}{\sqrt{L_s^2 + t_{\mathrm{ins}}^2}} \right) - \sqrt{2 \cdot t_{\mathrm{ins}}^2 + L_s^2} + \frac{t_{\mathrm{ins}}}{4} + \sqrt{L_s^2 + t_{\mathrm{ins}}^2} \right)$$

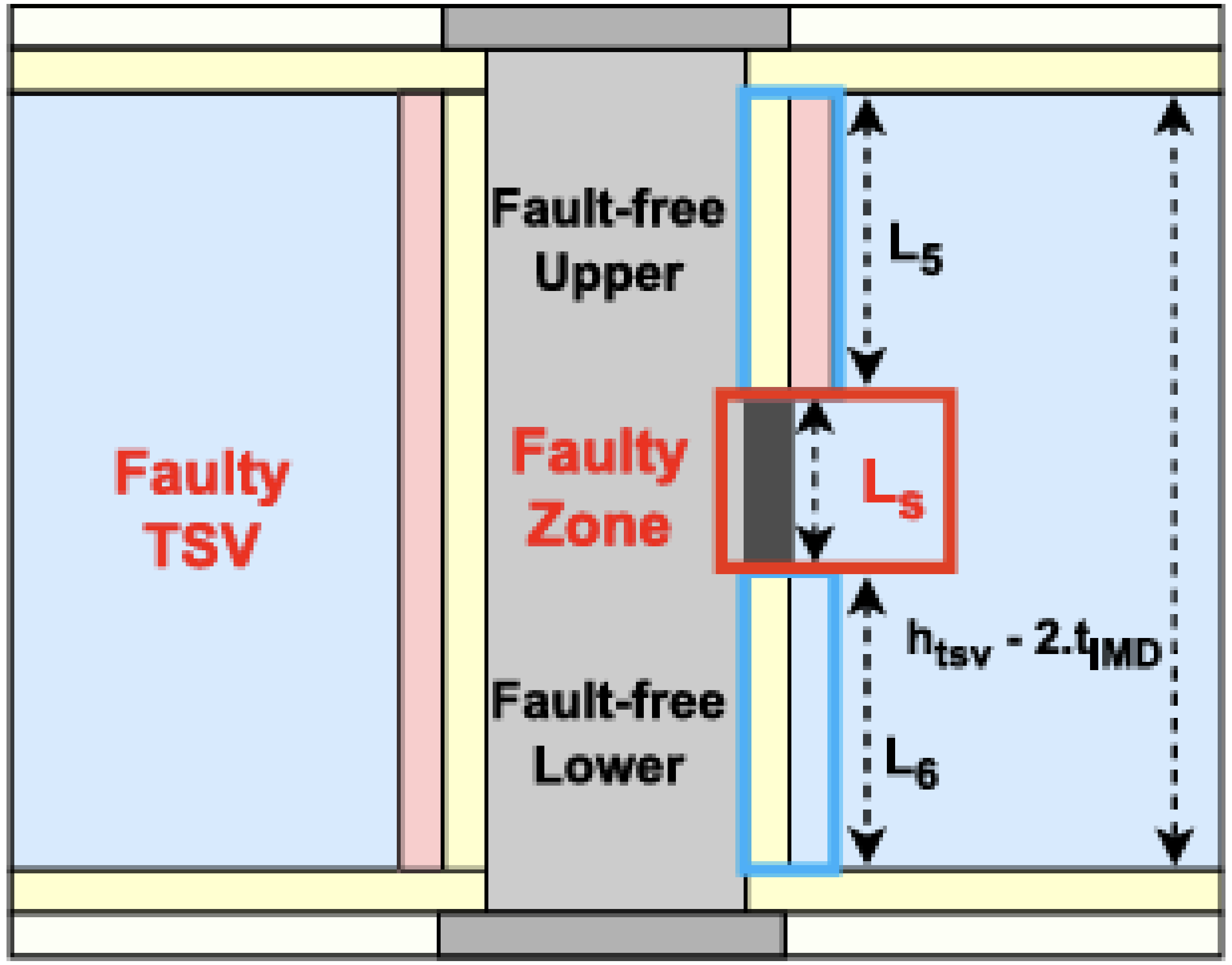
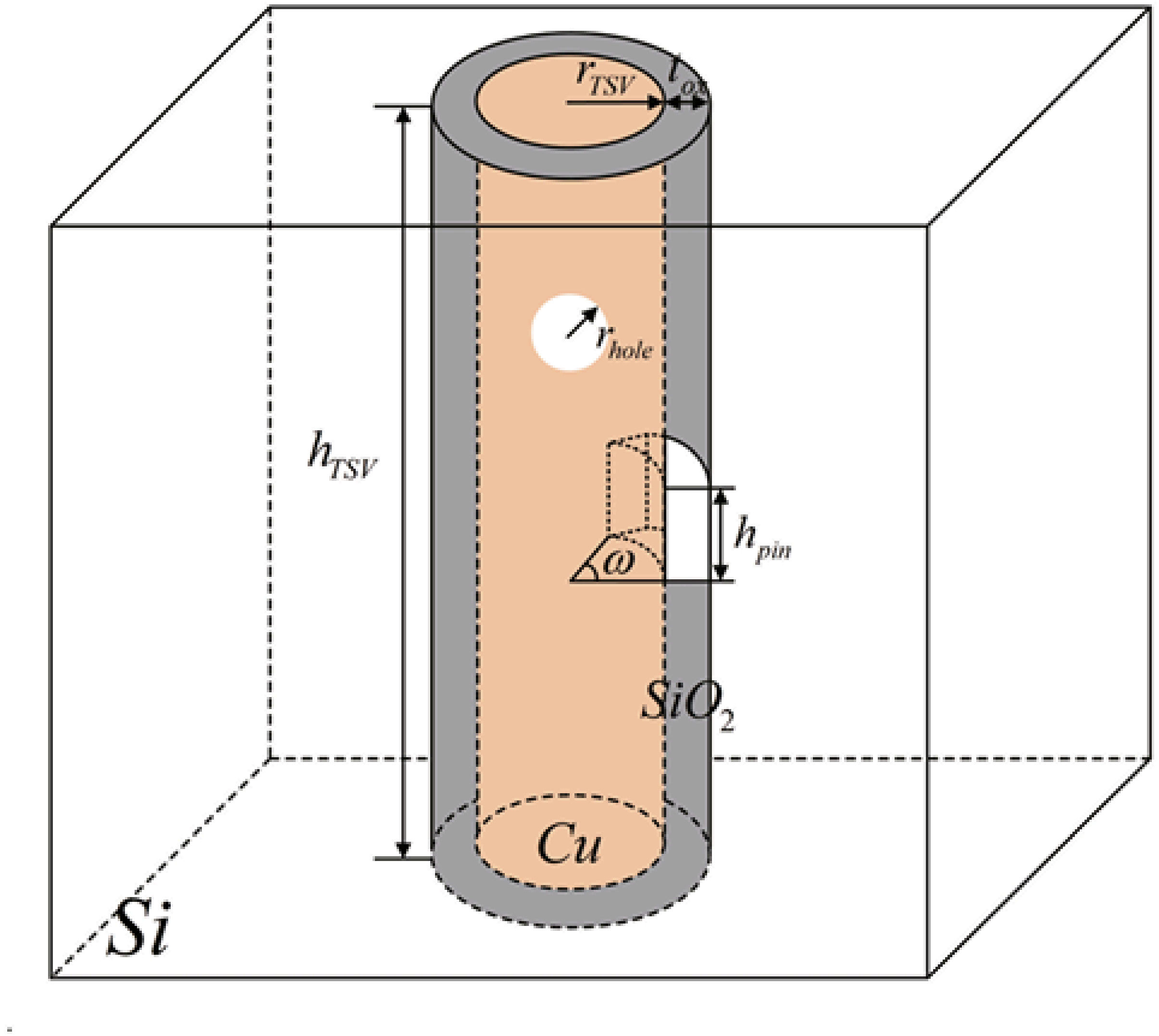


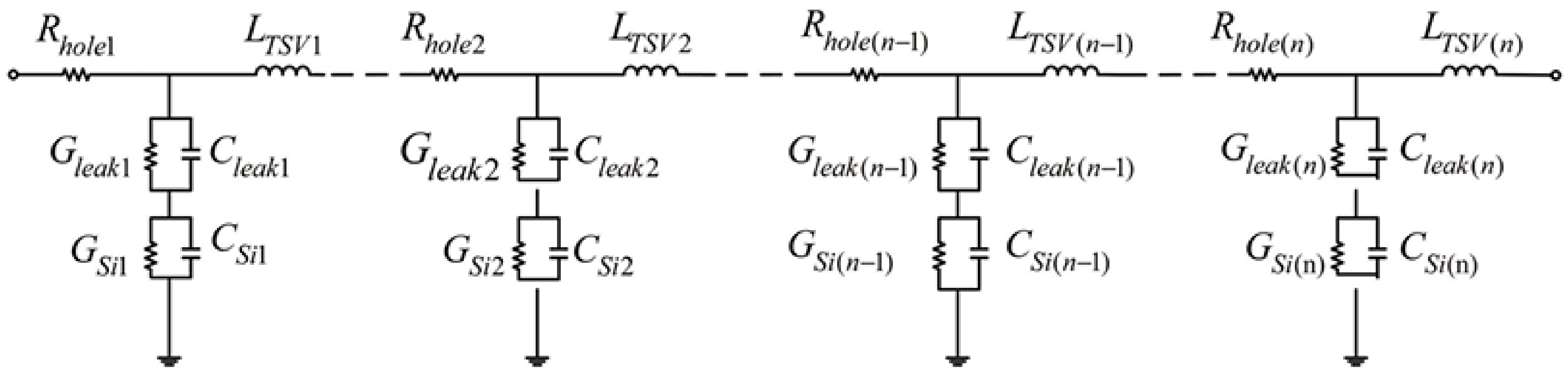
Fig.10.a. Schematic of TSV with short fault.(3)

Pinhole Defect in Terms of C_{Leak} and G_{Leak}



(a)

Fig.11.a. Defect TSV structure diagram[4]



(b)

Fig.11.b. defect TSV equivalent circuit diagram[4]

Equations of C_{Leak} and G_{Leak}

$$C_{leak} = \frac{2\pi\left(\varepsilon_0\varepsilon_{r,ox} l_{TSV} - \varepsilon_0\varepsilon_{r,air} j r_{TSV}\right)}{ln\left(\frac{r_{TSV}+t_{ox}}{r_{TSV}}\right)} [F]$$

Where $j = \dfrac{\omega h_{pin}}{2\pi r_{TSV}}$

$$\left[j = \frac{DamagedArea}{SideWallAreaAffected}\right]$$

$$C_{leak} = C_{ins} - C_{air}$$

$$G_{leak} = \frac{2\pi\sigma_{SiO_2} l_{TSV} - 4\pi^2 j f \varepsilon_0\varepsilon_{r,air} r_{TSV}}{\ln\left(\frac{r_{TSV}+t_{ox}}{r_{TSV}}\right)}$$

$$G_{leak} = G_{ox} - Y_{air}$$

G_{air} is almost zero so we take air Admittive capacitance

Workflow

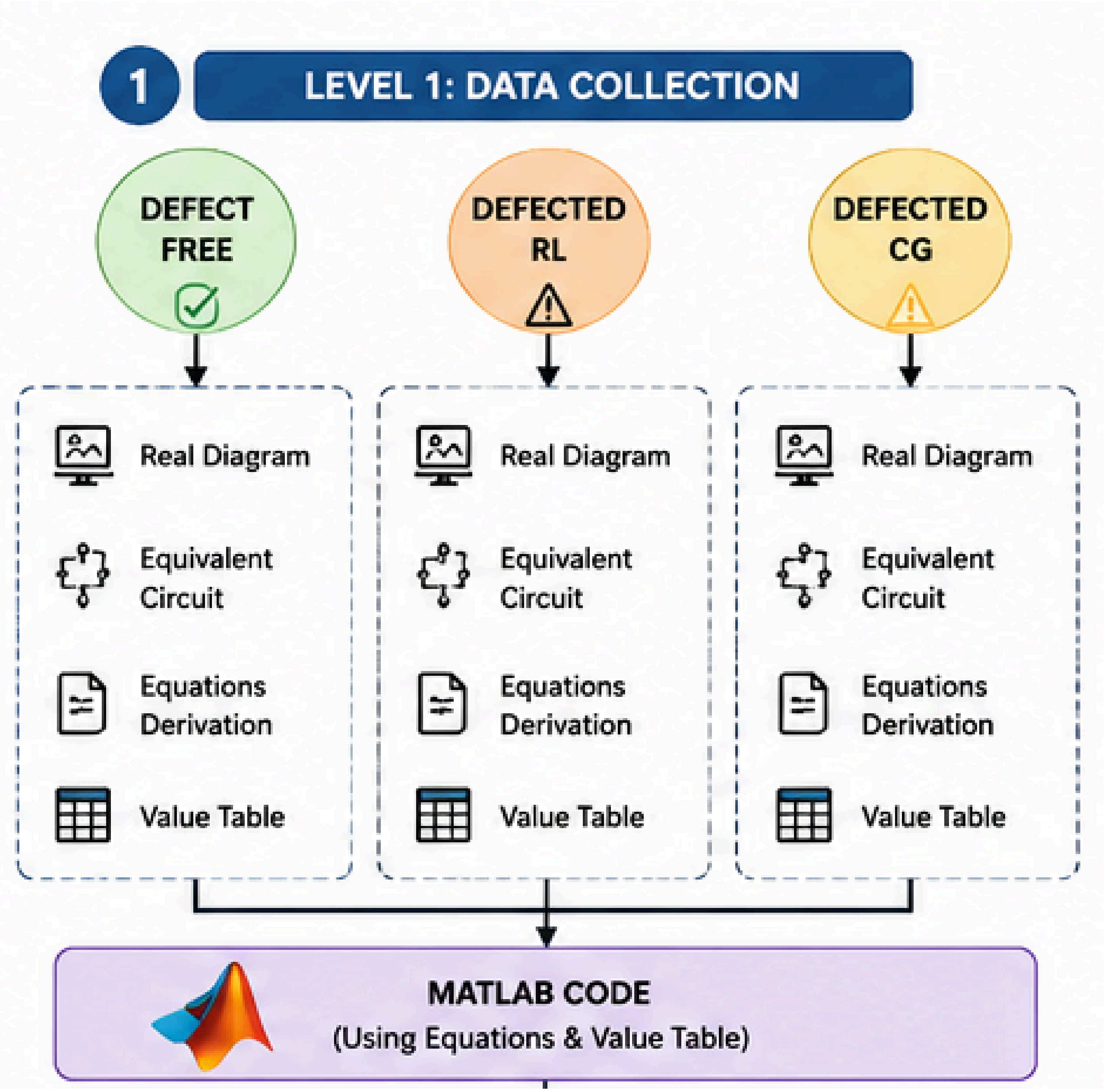


Fig.12.a. Level 01

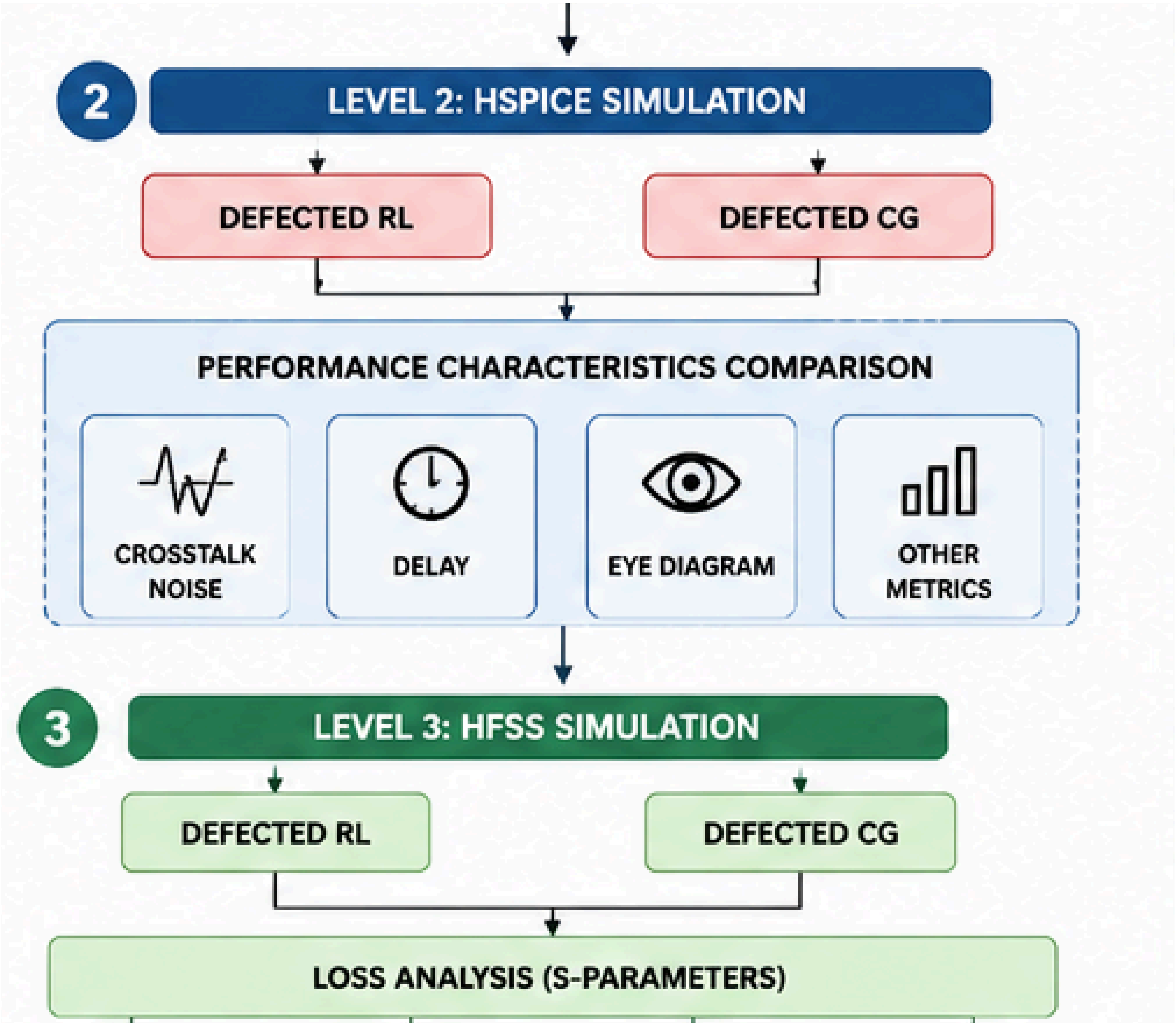


Fig.12.b. Level 02&03

References

- {1}Kim, J., Pak, J. S., Cho, J., Song, E., Cho, J., Kim, H., ... & Kim, J. (2011). High-frequency scalable electrical model and analysis of a through silicon via (TSV). IEEE Transactions on Components, Packaging and Manufacturing Technology, 1(2), 181-195.
- {2}Chandrakar, M., Solanki, K., Majumder, M. K., & Sharma, R. Reliability Concerns in Through Silicon Via based 3D Integration: Fabrication to Packaging.
- {3}Kirtonia, P., Williams, S., Akter, S., Bayoumi, M., & Khalil, K. (2025). A Novel TSV Model With Fault Characterization for High-Frequency Transmission in 3D ICs. IEEE Transactions on Circuits and Systems I: Regular Papers.
- {4}Liu, C., Dong, G., Zhi, C., & Zhu, Z. (2025). Modelling and simulation of TSV considering void and leakage defects. Journal of Computational Electronics, 24(2), 61.
- {5}Shang, Y., Tan, M., Li, C., & Sun, L. (2019). TSV manufacturing fault modelling and diagnosis based on multi-tone dither. Journal of Advanced Computational Intelligence and Intelligent Informatics, 23(1), 42-51.